Top-Down Pooling and Anticipatory Exposure

Abstract

The standard signaling story explains separation from below: low types cannot afford to imitate high types. This paper identifies the mirror failure mode in a two-type dynamic psychological signaling model. Sending an ambitious signal increases current persuasive value, but it also creates a one-period reputational exposure: today's action can be repriced tomorrow if the judgment state turns to high scrutiny.

Each type has a primitive self-model accuracy level, and a higher true type can carry higher exposure risk under adverse repricing. The high-scrutiny transition probability is $q = \Pr(j_{t+1} = H \mid j_t = L)$. The current rent from an ambitious signal for type θ is

$$D^L_\theta(q) = B_\theta - \beta q \mu_H \Phi_\theta.$$

The key is that $\Phi_H > \Phi_L$ and $B_H > 0$ can coexist.

This creates top-down collapse. Let

$$q_{\mathrm{IC}} = \frac{B_H}{\beta \mu_H \Phi_H}, \qquad q_{\mathrm{D1}} = \text{the D1 willingness threshold}, \qquad q^* = \max\{q_{\mathrm{IC}}, q_{\mathrm{D1}}\}.$$

For $q > q^*$, a binary-signal pure strategy that separates is not self-confirming: the high type would withdraw from s_A even on-path, and D1 then removes the high-type interpretation of off-path s_A.

The paper first proves a fixed-point selection result in the binary class: the unique D1-consistent assessment is pooling on the safe signal, while the ambitious signal becomes a signaling dead-end from above. A microfounded dynamic section then shows this collapse is robust to trust-capital and reinforcement effects: they shift margins, but bounded endogenous drift cannot generate automatic restoration. A continuous-type extension in Section 7 describes the observable shadow as polarization of ambitious senders, not simple mean reversion.

1 Introduction

In standard costly signaling, separation is sustained from below: low types cannot afford high-cost signaling, so they do not mimic high types. The model here studies the opposite failure mode. A good type may still benefit from an ambitious signal today, but can rationally retreat because that action creates a one-period exposure that can be repriced against her future credibility.

The mechanism is dynamic and belief-driven. The ambitious action has two effects. It is informative in the current period, but it also claims something about the sender's self-model. If high scrutiny arrives tomorrow, the claim can be reinterpreted as evidence of weaker self-understanding and thus impose an additional penalty. This is adverse repricing, not a true-type audit.

The primitives are minimal: two types, two sender signals, and an exogenous binary judgment state. The sender observes the low-scrutiny risk $q = \Pr(j_{t+1} = H \mid j_t = L)$ before acting. If she sends the ambitious signal, tomorrow's state can impose an extra epistemic cost. If she sends the pooling-safe action, that one-period shadow disappears.

The model has three moving parts.

1. A static, one-period D1 core that shows a top-down reversal: high types exit separation first when q is large.
2. A fixed-point closure argument that combines sender ICs and D1 beliefs in a binary-signal pure-strategy psychological assessment.
3. A dynamic forward-invariance result that adds trust-capital/semantic dynamics and shows endogenous accumulation does not automatically repair the collapse.

The first stage identifies how adverse repricing plus a D1-style off-path map can select pooling from above. The second stage closes the belief loop in binary form: a self-referential interpretation where high occupancy of the ambitious signal implies high-type meaning, but then current IC may already fail, and off-path reinterpretation then collapses that meaning. The third stage asks whether time and dynamics can reverse this outcome.

The result is defensive: no new stopping phenomenon appears. Static and belief-loop collapse does not self-correct through learning or accumulation alone.

Section 2 introduces primitives and the one-period IC comparison. Section 3 derives the exposure ranking from primitive self-accuracy levels. Section 4 gives the static D1 core and the fixed-point threshold $q^* = \max\{q_{\mathrm{IC}}, q_{\mathrm{D1}}\}$. Section 5 closes the belief loop in binary strategies. Section 6 gives the dynamic forward-invariance result. Section 7 presents a continuous-type extension, where the surviving ambitious signal can become polarized.

2 Primal Structure and Benchmark

This paper studies a signaling environment with two sender types and two signals, where the same signal that can separate today also creates a near-term reputation risk that may be repriced tomorrow. The baseline is the dynamic extension of Spence logic needed for the fixed-point reversal.

2.1 Types, signals, and judgment states

There is a sender with type

$$\theta \in \{\theta_L, \theta_H\}, \qquad \theta_H > \theta_L.$$

The receiver does not observe the type.

The sender chooses one of two signals

$$s \in \{s_A, s_P\}.$$

Signal s_A is the ambitious action, and s_P is the pooling-safe action.

Judgment intensity follows a public Markov state

$$j_t \in \{L, H\}.$$

State L is ordinary evaluation, and state H is high-scrutiny evaluation. Let

$$\mu_H > \mu_L > 0$$

denote the scrutiny multipliers and

$$q := \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability from low to high. The sender observes q before choosing a signal.

2.2 One-period exposure shadow

Choosing s_A creates a one-period exposure state that can be repriced tomorrow. Let

$$X_t \in \{0, 1\}$$

indicate whether last period's action is exposed in the next period. If $s_t = s_A$, then $X_{t+1} = 1$; if $s_t = s_P$, then $X_{t+1} = 0$. In this baseline the exposure horizon is one period: choosing the safe signal today removes the next-period exposure created by a yesterday ambitious action.

If $X_{t+1} = 1$ and $j_{t+1} = H$, the sender incurs a future epistemic penalty proportional to a true-type exposure index. This is the anticipatory shadow that drives behavior in the low state.

2.3 Current payoff and the anticipatory rent

Let B_θ denote the current net separation surplus from s_A versus s_P in state L when the receiver interprets the ambitious signal in the high-type way. The low-state payoff gain from sending s_A is

$$D^L_\theta(q) = B_\theta - \beta q \mu_H \Phi_\theta,$$

where $\Phi_\theta > 0$ is the true-type exposure cost derived from primitive self-model accuracy (Section 3), and $\beta \in (0, 1)$ is the discount factor.

The type-θ IC threshold is

$$q^*_\theta := \frac{B_\theta}{\beta \mu_H \Phi_\theta},$$

whenever $B_\theta > 0$. Type θ is willing to send s_A in the low-scrutiny state iff

$$q < q^*_\theta.$$

2.4 Static benchmark

When $q = 0$, current risk from future repricing is absent. The model then collapses to a one-shot separating benchmark.

Assumption 1 (Static separation)**.**

$$B_H > 0, \qquad B_L < 0.$$

Under Assumption 1, with no anticipated repricing s_A is uniquely attractive only to the high type.

2.5 Anticipatory reversal object

The model's distinctive case is high-type exposure vulnerability. Even if $B_H > 0$, the same transition probability q can make high types abandon s_A because high types lose more from possible repricing in state H.

As the baseline, define

$$q_{\mathrm{IC}} := \frac{B_H}{\beta \mu_H \Phi_H},$$

which is the IC threshold that would bind if s_A were still interpreted as a high-type signal.

Define also the D1-type willingness threshold

$$q_{\mathrm{D1}} > 0$$

from the off-path comparison in Section 4.

The combined threshold is

$$q^* := \max\{q_{\mathrm{IC}}, q_{\mathrm{D1}}\}.$$

For $q > q^*$, the top-down reversal can be pinned down in static one-period terms: high types lose the static incentive to stay on s_A, and D1 no longer keeps s_A high-type in off-path beliefs.

3 Psychological Exposure and Bayesian Surprise

The benchmark environment in Section 2 treats the ambitious signal s_A as a costly signal of ability. This section adds the psychological object that distinguishes the model from standard costly signaling. The ambitious signal does not only reveal information about the sender's type. It also exposes the sender's self-model. A sender who chooses s_A represents herself as someone whose private assessment of her own capacity is accurate enough to justify taking the ambitious action. The social cost of later reinterpretation is therefore not only a material or reputational loss. It is also an epistemic loss: observers revise their beliefs about whether the sender correctly understood her own type.

The section makes three clarifications. First, the exposure index Φ_θ is a true-type-indexed primitive object, derived from primitive self-model-accuracy levels. It is not an equilibrium object generated by a previous separating history. Second, the relevant channel is adverse repricing, not truth verification. A high-precision audit would tend to confirm genuinely accurate high types; that is not the technology studied here. Third, the D1 argument later uses Φ_θ as a cost attached to the sender's true type, not as a cost attached to the receiver's off-path attribution. This keeps the exposure technology separate from the belief refinement that interprets deviations.

3.1 Belief-dependent exposure

Let $j \in \{L, H\}$ denote the current judgment-intensity state, with associated scrutiny multiplier $\mu_j > 0$. The low state L represents ordinary evaluation and the high state H represents intensified evaluation. The state variable j is not a political primitive. It is an exogenous state of the social evaluation technology.

After s_A, observers form beliefs not only about the sender's ability θ, but also about the accuracy of the sender's self-model. Write

$$m \in \{0, 1\}$$

for the event that the sender's self-model is accurate, where $m = 1$ denotes accurate self-assessment. The sender cares not only about what observers infer about θ, but also about what observers infer about the sender's own assessment of θ.

For a finite hierarchy of beliefs $(\mu^2, \dots, \mu^k)$, let

$$L(\mu^2, \dots, \mu^k; j, x)$$

denote the epistemic penalty in judgment state j and exposure state $x \in \{0, 1\}$. The exposure state records whether the sender previously chose the ambitious signal. Choosing s_A sets $x = 1$ next period; choosing s_P clears exposure next period.

The total one-period psychological payoff can be written as

$$u_\theta(s, j, x; \mu^2, \dots, \mu^k) = v_\theta(s) - c_\theta(s) - \mu_j L(\mu^2, \dots, \mu^k; j, x),$$

where $v_\theta(s)$ is the ordinary reputational or material payoff from the signal, $c_\theta(s)$ is the intrinsic cost of the signal, and the final term is the judgment-weighted epistemic penalty.

The next assumption is purely regularity. It rules out discontinuities inside a fixed judgment state. The discontinuities studied below will come from changes in the transition probability into high scrutiny, not from discontinuous psychological utility within a regime.

Assumption 2 (Regime-conditional continuity)**.** *For each fixed $j \in \{L, H\}$ and $x \in \{0, 1\}$, the function*

$$L(\mu^2, \ldots, \mu^k; j, x)$$

is continuous in the relevant finite hierarchy of higher-order beliefs. The action set, type space, and exposure state space are finite.

Lemma 1 (Regime-conditional continuity)**.** *Under Assumption 2, for each fixed judgment-intensity state j, the induced psychological payoff is continuous in the relevant higher-order belief hierarchy. Hence the regime-conditional psychological game satisfies the standard continuity requirement for belief-dependent payoffs. On any open region where the relevant incentive constraints are strict, the associated pure-strategy equilibrium selection is locally constant.*

Proof. Fix j and x. The ordinary payoff terms $v_\theta(s)$ and $c_\theta(s)$ are defined on a finite action and type space. The multiplier μ_j is constant within the state j. By Assumption 2, the epistemic penalty $L(\mu^2, \ldots, \mu^k; j, x)$ is continuous in the relevant belief hierarchy. Multiplication by the fixed scalar μ_j and addition of finite ordinary payoff terms preserve continuity. Thus, the regime-conditional psychological payoff is continuous.

Since the type and action spaces are finite, strict incentive inequalities are preserved under sufficiently small perturbations of beliefs and payoff parameters within a fixed state. Therefore, wherever the relevant incentive constraints are strict, the same pure-strategy profile remains selected in a neighborhood of the primitive. Any discontinuous change in equilibrium behavior must therefore come from a change in the relevant state, transition probability, or binding incentive constraint, rather than from a discontinuity of belief-dependent utility within the fixed state. □

3.2 Primitive self-accuracy levels

The exposure index is built from a level channel, not a precision channel. Let

$$p_\theta := \Pr(m = 1 \mid \theta)$$

denote the primitive probability that a type-θ sender has an accurate self-model. The key ordering is

$$1 - \delta > p_H > p_L > \bar{p} > \delta > 0.$$

Higher types are genuinely more self-accurate in the primitive environment. The lower benchmark $\bar{p}$ is the assessment induced by adverse repricing under high scrutiny.

This assumption should not be read as signal precision. Precision is a fidelity concept: a more precise audit reveals the true state more accurately. If high types are genuinely more accurate, a fidelity-improving audit tends to confirm them, not punish them. The model instead studies adverse social repricing. In a high-scrutiny state, an exposed self-presentation can be reinterpreted downward with a positive probability that is not driven by the sender's true merit. The economic loss is increasing in the amount of self-model reputation-at-risk carried by the true type.

Let $\alpha \in (0, 1]$ be the probability, conditional on high scrutiny and exposure, that the ambitious self-presentation is adversely repriced. This probability is taken to be independent of type in the baseline. It can be normalized into μ_H, but keeping it visible makes the interpretation clear: high scrutiny is not a deterministic truth audit. It is a state in which exposed self-presentations face a type-independent risk of adverse reinterpretation.

The epistemic exposure index is

$$\Phi_\theta = \alpha\, D_{\mathrm{KL}}\left(\mathrm{Bern}(p_\theta) \,\|\, \mathrm{Bern}(\bar{p})\right).$$

The direction of the divergence reflects the economic object. The loss is not the ordinary probability of failure. It is the collapse from a carried self-accuracy level to a lower adverse benchmark. Judgment intensity prices this exposure:

$$C_E(\theta, j) = \mu_j \Phi_\theta.$$

Thus Φ_θ is neither history-induced reputation capital nor signal precision. It is the true-type-indexed loss from adverse repricing of self-model exposure. The receiver's later off-path attribution may affect the reputational return to a deviation, but it does not redefine Φ_θ.

3.3 Bayesian-surprise complementarity

Lemma 2 (Bayesian-surprise foundation for scrutiny-type complementarity)**.** *Suppose*

$$1 - \delta > p_H > p_L > \bar{p} > \delta > 0.$$

Define

$$\Phi_\theta = \alpha\, D_{\mathrm{KL}}\left(\mathrm{Bern}(p_\theta) \,\|\, \mathrm{Bern}(\bar{p})\right)$$

with $\alpha > 0$*, and let*

$$C_E(\theta, j) = \mu_j \Phi_\theta.$$

Then the epistemic cost has increasing differences in type and judgment intensity. For every $\mu' > \mu$*,*

$$C_E(\theta_H, \mu') - C_E(\theta_H, \mu) > C_E(\theta_L, \mu') - C_E(\theta_L, \mu).$$

Equivalently, the marginal cost of higher judgment intensity is larger for the high-self-accuracy type.

Proof. For Bernoulli beliefs,

$$D_{\mathrm{KL}}\left(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(\bar{p})\right) = p \log \frac{p}{\bar{p}} + (1-p) \log \frac{1-p}{1-\bar{p}}.$$

Differentiating with respect to p gives

$$\frac{\partial}{\partial p} D_{\mathrm{KL}}\left(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(\bar{p})\right) = \log \frac{p(1-\bar{p})}{\bar{p}(1-p)}.$$

This derivative is positive whenever $p > \bar{p}$. Since $p_H > p_L > \bar{p}$, it follows that

$$\Phi_H > \Phi_L.$$

Therefore, for every $\mu' > \mu$,

$$\left[C_E(\theta_H, \mu') - C_E(\theta_H, \mu)\right] - \left[C_E(\theta_L, \mu') - C_E(\theta_L, \mu)\right] \qquad = (\mu' - \mu)(\Phi_H - \Phi_L) > 0.$$

Thus higher judgment intensity raises the epistemic cost of the high-self-accuracy type more than that of the low-self-accuracy type. □

Lemma 2 is the first reversal relative to standard signaling theory. In a Spence environment, high types are more willing to bear the ambitious signal because they face a lower ordinary cost of sending it. Here, high types also carry more self-model reputation-at-risk. When high scrutiny adversely reprices exposed self-presentation, the same type characteristic that supports ambitious action under low scrutiny becomes the source of larger exposure.

Remark 1 (Repricing, not precision)**.** *The sign of Lemma 2 depends on the level channel. Replacing p_θ with signal precision would change the model. A higher-precision audit is a fidelity technology and tends to confirm an accurate high type. It therefore does not generate top-down withdrawal. The primitive needed here is a type-indexed self-accuracy level combined with an adverse repricing state, not a variance or precision parameter.*

3.4 From self-accuracy to crash risk

The preceding lemma converts primitive self-accuracy into exposure. A high type has more to gain from separation, but also more to lose from a future state that adversely reinterprets her ambitious signal as evidence about self-model risk.

Let B_θ denote the current net separation rent from choosing s_A rather than s_P in the low-judgment state, before future high-scrutiny exposure is priced. Let

$$q = \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability into high scrutiny. If choosing s_A creates the one-period shadow, the low-state payoff gain from choosing s_A can be written as

$$D_\theta^L(q) = B_\theta - \beta q \mu_H \Phi_\theta.$$

The associated threshold is

$$q_\theta^* = \frac{B_\theta}{\beta \mu_H \Phi_\theta},$$

whenever $B_\theta > 0$.

Thus top-down withdrawal is governed not by exposure alone, but by the ratio between exposure and current rent:

$$\frac{\Phi_\theta}{B_\theta}.$$

The high type exits first precisely when

$$\frac{\Phi_H}{B_H} > \frac{\Phi_L}{B_L}.$$

In that case,

$$q_H^* < q_L^*.$$

As the probability of future high scrutiny increases, the high type's incentive constraint binds first.

Corollary 1 (Top-down threshold ordering)**.** *Suppose $B_H > 0$, $B_L > 0$, and*

$$\frac{\Phi_H}{B_H} > \frac{\Phi_L}{B_L}.$$

Then

$$q_H^* < q_L^*.$$

Hence an increase in the transition probability into high scrutiny first destroys the ambitious-signal incentive of the type with the larger exposure-to-rent ratio.

Proof. The threshold is

$$q_\theta^* = \frac{B_\theta}{\beta \mu_H \Phi_\theta}.$$

Since $\beta\mu_H > 0$,

$$q_H^* < q_L^*$$

iff and only if

$$\frac{B_H}{\Phi_H} < \frac{B_L}{\Phi_L},$$

which is equivalent to

$$\frac{\Phi_H}{B_H} > \frac{\Phi_L}{B_L}.$$

□

The corollary gives the transition from the psychological exposure mechanism to the dynamic pooling result. The high type is not less capable of sending the ambitious signal. She is more exposed because her true type carries more self-model reputation-at-risk under adverse repricing.

4 Anticipatory Exposure and the D1 Flip

The previous section established two ingredients. First, conditional on a fixed judgment-intensity state, the belief-dependent payoff is continuous in the relevant higher-order belief hierarchy. Hence the psychological-game object is well-defined within each regime slice. Second, the epistemic exposure term is pinned to the sender's true type through primitive self-accuracy levels and adverse repricing. It is not a signal-precision parameter and not an equilibrium posterior.

This section shows how these ingredients change equilibrium selection. In standard costly signaling, the ambitious signal is credible because high types are most willing to send it. Here, sufficiently likely future scrutiny reverses that willingness ranking. The high type becomes less willing than the low type to carry the ambitious signal, because the high type has more self-model reputation-at-risk under adverse repricing. This reversal has two consequences. First, the separating profile fails when the high type's anticipatory incentive constraint binds. Second, in the pooling candidate, a D1-style forward-induction refinement no longer interprets an off-path ambitious signal as evidence of the high type.

4.1 The anticipatory incentive constraint

Let the current judgment-intensity state be low, denoted L. Let

$$q = \Pr(j_{t+1} = H \mid j_t = L)$$

be the transition probability into the high-scrutiny state. The agent chooses between the ambitious signal s_A and the pooling-safe signal s_P. Sending s_A creates an exposure state that can be repriced if the next-period state is H.

For type $\theta \in \{\theta_L, \theta_H\}$, let B_θ denote the current net separation surplus from sending s_A rather than s_P, before future high-scrutiny exposure is priced. This term includes the current reputational benefit of being interpreted as high type and subtracts current intrinsic and low-scrutiny epistemic costs. Let $\Phi_\theta > 0$ denote the true-type-indexed adverse-repricing exposure cost derived in Section 3. The low-state payoff gain from sending s_A is

$$D_\theta(q) = B_\theta - \beta q \mu_H \Phi_\theta,$$

where $\beta \in (0, 1)$ is the discount factor and μ_H is the high-scrutiny intensity.

The high type's separating incentive constraint is

$$D_H(q) \geq 0.$$

Thus define

$$q_{\mathrm{IC}} = \frac{B_H}{\beta \mu_H \Phi_H},$$

assuming $B_H > 0$. For $q > q_{\mathrm{IC}}$, the high type strictly prefers the pooling-safe signal s_P to remaining on the ambitious signal s_A, even if s_A is still interpreted as high type.

Lemma 3 (Anticipatory failure of separation)**.** *Suppose the current state is L. If*

$$q > q_{\mathrm{IC}},$$

then the separating profile

$$\theta_H \mapsto s_A, \qquad \theta_L \mapsto s_P$$

is not a psychological equilibrium.

Proof. Under the separating profile, s_A is interpreted as the high-type signal. Hence the high type receives the current separation surplus B_H from s_A, but also internalizes the future exposure cost $\beta q \mu_H \Phi_H$. Its payoff gain from choosing s_A rather than s_P is therefore

$$D_H(q) = B_H - \beta q \mu_H \Phi_H.$$

If $q > q_{\mathrm{IC}}$, then $D_H(q) < 0$. Thus the high type strictly prefers s_P to s_A. The proposed separating profile violates the high type's incentive constraint and therefore cannot be an equilibrium. □

The point is not that the low type successfully mimics the high type. The separating profile collapses before that question arises. The high type exits because the expected future cost of adverse repricing exceeds the current benefit of being recognized.

4.2 The D1 flip

It remains to check whether pooling on s_P can be sustained. The possible obstruction is off-path resurrection: if all types pool on s_P, an off-path deviation to s_A might again be interpreted as evidence of high type, restoring the value of the ambitious signal.

This is where the refinement matters. Consider the pooling candidate in which both types choose s_P. The receiver observes an off-path signal s_A. Let a denote a receiver response, and let $R(a)$ be the reputational return from that response. For simplicity, assume $R(a)$ is type-independent and ordered by receiver favorability.

Let the cost of deviating to the ambitious signal for type θ be

$$K_\theta(q) = \bar{k}_\theta + \beta q \mu_H \Phi_\theta,$$

where $\bar{k}_\theta$ is the current cost of sending s_A in the pooling environment and $\beta q \mu_H \Phi_\theta$ is the anticipatory high-scrutiny exposure cost.

The exposure term Φ_θ is the deviator's true-type cost. D1 determines how the receiver interprets the off-path signal and therefore affects $R(a)$. It does not relabel the deviator's exposure technology. This separation is essential: otherwise the off-path belief used by the refinement would also redefine the cost object being compared.

Given receiver response a, type θ's payoff gain from deviating to s_A is

$$G_\theta(a; q) = R(a) - K_\theta(q).$$

Define the profitable-deviation response set

$$\mathcal{D}_\theta(q) = \{a : G_\theta(a; q) > 0\}.$$

In the standard Spence ranking, the high type has a lower signaling cost, so

$$\mathcal{D}_L(q) \subset \mathcal{D}_H(q).$$

D1 then assigns the off-path ambitious signal to the high type. The present model reverses this inclusion when future scrutiny is sufficiently likely. By Lemma 2, high types have larger future exposure:

$$\Phi_H > \Phi_L.$$

Hence the high type's cost increases faster in q. The ranking flips when

$$K_H(q) > K_L(q),$$

or equivalently,

$$\bar{k}_H + \beta q \mu_H \Phi_H > \bar{k}_L + \beta q \mu_H \Phi_L.$$

Define

$$q_{\mathrm{D1}} = \frac{\bar{k}_L - \bar{k}_H}{\beta \mu_H (\Phi_H - \Phi_L)},$$

whenever $\bar{k}_L > \bar{k}_H$. Then for $q > q_{\mathrm{D1}}$,

$$K_H(q) > K_L(q).$$

Lemma 4 (D1 flip)**.** *Suppose $\Phi_H > \Phi_L$ and $q > q_{\mathrm{D1}}$. Then*

$$\mathcal{D}_H(q) \subsetneq \mathcal{D}_L(q).$$

Therefore, under the D1 criterion, an off-path ambitious signal s_A cannot be assigned to the high type. It must be assigned to the low type.

Proof. For any receiver response a, type θ's gain from deviating to s_A is

$$G_\theta(a; q) = R(a) - K_\theta(q).$$

If $q > q_{\mathrm{D1}}$, then $K_H(q) > K_L(q)$. Hence for every receiver response a,

$$G_H(a; q) < G_L(a; q).$$

Therefore, whenever the deviation is profitable for the high type, it is also profitable for the low type:

$$G_H(a; q) > 0 \quad \Rightarrow \quad G_L(a; q) > 0.$$

Thus

$$\mathcal{D}_H(q) \subseteq \mathcal{D}_L(q).$$

The inclusion is strict whenever there exists a receiver response a such that

$$K_L(q) < R(a) < K_H(q).$$

Under D1, if one type can profitably deviate under a strictly wider set of receiver responses, the deviation is attributed to that type rather than to the type whose profitable-deviation set is smaller. Hence the off-path signal s_A is attributed to θ_L, not to θ_H. □

The ambitious signal therefore changes its meaning. In the standard model, an off-path ambitious signal says: this must be a strong type, because only strong types can afford it. Here, when future scrutiny is sufficiently likely, the same signal says: this is unlikely to be the high type, because high types have the most to lose from carrying exposed ambition into future scrutiny.

4.3 Top-down pooling

We can now combine the two thresholds. The first threshold kills the separating profile by violating the high type's incentive constraint. The second threshold kills the off-path resurrection of the ambitious signal under D1.

Define

$$q^* = \max\{q_{\mathrm{IC}}, q_{\mathrm{D1}}\}.$$

Theorem 1 (Top-down pooling)**.** *Suppose:*

1. *$B_H > 0$, so the high type would choose the ambitious signal absent anticipatory high-scrutiny exposure;*
2. *$\Phi_H > \Phi_L$, as derived from primitive self-accuracy levels and Bayesian-surprise repricing in Section 3;*
3. *$q > q^* = \max\{q_{\mathrm{IC}}, q_{\mathrm{D1}}\}$;*
4. *the receiver's off-path beliefs satisfy the D1 criterion;*
5. *the analysis is restricted to the binary-signal pure-strategy class $\{s_A, s_P\}$.*

Then the unique D1-consistent pure-strategy psychological equilibrium in this class is pooling on the safe signal:

$$\theta_H \mapsto s_P, \qquad \theta_L \mapsto s_P.$$

Proof. First, since $q > q_{\mathrm{IC}}$, Lemma 3 implies that the separating profile cannot be an equilibrium. Even if s_A is interpreted as the high-type signal, the high type strictly prefers s_P, because its expected future exposure cost exceeds its current separation surplus.

Second, consider the pooling candidate in which both types choose s_P. Since $q > q_{\mathrm{D1}}$, Lemma 4 implies that the off-path ambitious signal s_A is attributed by D1 to the low type rather than to the high type. Therefore s_A carries no high-type reputational premium under D1-consistent beliefs.

Given this off-path belief, the high type has no profitable deviation to s_A. The deviation would create future exposure while failing to restore the high-type interpretation. The low type also has no profitable deviation: under the D1-consistent posterior, its deviation is interpreted as low type and therefore does not generate the reputational return needed to cover the cost of s_A.

Thus both types optimally choose s_P. Since separation has already been ruled out by the high type's incentive constraint, and the only off-path resurrection of s_A is ruled out by D1, the pooling profile is the unique D1-consistent pure-strategy psychological equilibrium within the binary-signal class. □

The theorem identifies a reversal of the usual signaling logic. In standard costly signaling, separation fails from below when low types become able to mimic high types. Here, separation fails from above. The high type exits first because it has both the current separation rent and the largest future adverse-repricing liability.

Equivalently, the ambitious signal loses its credibility not because it becomes cheap for low types, but because it becomes too exposed for high types. Once future scrutiny is sufficiently likely, an agent who still sends the ambitious signal is no longer interpreted as confident. Under D1, it is interpreted as failing to internalize the future cost of exposure.

4.4 Interpretation

The result can be summarized by contrasting two mechanisms. In the standard Spence model, bad types cannot climb up. The high-cost signal is credible because only high types are willing to bear it. In the present model, good types climb down. The ambitious signal becomes dangerous precisely for the types that previously used it to separate.

This produces top-down pooling. The pooling outcome is not generated by low-type imitation. It is generated by high-type withdrawal. The signal s_A does not merely lose precision; its off-path meaning reverses. After the D1 flip, s_A is no longer evidence of high ability. It is evidence of low anticipatory discipline.

This is the belief-driven pooling reversal. Future scrutiny changes the sender's current incentives, and D1 changes the receiver's interpretation of deviations. Together they close the fixed point: high types withdraw because future scrutiny makes ambitious exposure costly, and once high types are expected to withdraw, ambitious deviations cease to be high-type credible.

5 Belief Loops and Fixed-Point Selection

Section 4 gives the one-period D1 logic. This section closes that logic into a fixed point over psychological assessment. The fixed point is not only over sender strategies, but also over the receiver's interpretation of the ambitious signal and the sender's expectation of that interpretation.

5.1 Psychological assessment

At a low-scrutiny history, let

$$\sigma_\theta \in \{s_A, s_P\}$$

derive the sender's strategy of type θ. Let $\lambda_A \in [0,1]$ be the receiver's posterior on type θ_H after observing s_A. If s_A is on-path, λ_A is pinned down by Bayes' rule; if s_A is off-path, λ_A is selected through D1.

Let

$$B_H(\lambda)$$

denote the type-H sender's current net benefit from choosing s_A rather than s_P when the ambitious signal is interpreted with posterior λ. Assume

$$B_H(\lambda) \text{ is weakly increasing in } \lambda, \qquad B_H(1) > 0, \qquad B_H(0) \le 0,$$

and let

$$B_L(0) < 0.$$

The first inequality means high-type current return rises with favorable high attribution; the last two say an ambitious signal read as low-type evidence is not profitable for either type.

Future repricing risk is true-type indexed as in Section 3; $\Phi_H > \Phi_L > 0$. For receiver posterior λ, the low-state net advantage of staying ambitious is

$$\Delta_H(\lambda, q) = B_H(\lambda) - \beta q \mu_H \Phi_H.$$

This gives the same in-spot incentive threshold

$$q_{\mathrm{IC}} := \frac{B_H(1)}{\beta \mu_H \Phi_H}.$$

5.2 Two candidate beliefs

Let

$$z \in \{0, 1\}$$

index whether the high type is expected to occupy s_A.

When $z = 1$ ("stay" candidate), both strategy and interpretation are separating:

$$\lambda_A(1) = 1.$$

When $z = 0$ ("withdraw/pooling" candidate), s_A is off-path. Under D1 and $q > q_{\mathrm{D1}}$, low-type deviating opportunities dominate at any receiver response, so the off-path ambitious action is attributed to low type and

$$\lambda_A(0) = 0.$$

Here

$$q_{\mathrm{D1}} := \inf \left\{ q : \ \bar{k}_H + \beta q \mu_H \Phi_H > \bar{k}_L + \beta q \mu_H \Phi_L \right\},$$

where $\bar{k}_\theta$ is the current off-path action cost part, as in Section 4. This is the D1 threshold from the profitable-deviation inclusion comparison.

The belief-loop operator is

$$T_q(z) := \mathbf{1}\{\Delta_H(\lambda_A(z), q) \geq 0\}.$$

A psychological assessment is a fixed point if $z = T_q(z)$.

5.3 Fixed-point selection

Theorem 2 (Binary belief-loop fixed point)**.** *Suppose:*

1. *$B_H(\lambda)$ is weakly increasing with $B_H(1) > 0$ and $B_H(0) \leq 0$;*
2. *$B_L(0) < 0$;*
3. *$\Phi_H > \Phi_L$;*
4. *$q > q^* = \max\{q_{\mathrm{IC}}, q_{\mathrm{D1}}\}$;*
5. *off-path beliefs satisfy D1;*
6. *the analysis is restricted to the binary-signal pure-strategy class.*

Then the unique psychological assessment in this class has

$$\sigma_H = s_P, \qquad \sigma_L = s_P.$$

Equivalently, the unique fixed point is $z^* = 0$.

Proof. For $z = 1$, the receiver is induced to read s_A as high type, so $\Delta_H(\lambda_A(1), q) = \Delta_H(1, q) = B_H(1) - \beta q \mu_H \Phi_H < 0$ by $q > q_{\mathrm{IC}}$. Therefore $T_q(1) = 0$, so $z = 1$ is not a fixed point.

For $z = 0$, D1 gives $\lambda_A(0) = 0$, and $q > q_{\mathrm{D1}}$. Hence

$$\Delta_H(\lambda_A(0), q) = \Delta_H(0, q) = B_H(0) - \beta q \mu_H \Phi_H < 0,$$

using $B_H(0) \leq 0$ and $\beta q \mu_H \Phi_H > 0$. So $T_q(0) = 0$, so $z = 0$ is a fixed point.

Any profile where only the low type sends s_A has $\lambda_A = 0$ on path. That profile is not profitable for low type because $B_L(0) < 0$.

Any profile with both types sending s_A has $\lambda_A = \pi_0 \in (0, 1)$. Since $B_H(\lambda)$ is weakly increasing,

$$\Delta_H(\pi_0, q) \le \Delta_H(1, q) < 0,$$

so high type deviates to s_P when $q > q_{\text{IC}}$. Hence pooling on s_A is not an equilibrium.

The only remaining binary-strategy profile is pooling on s_P, and both types' profitable-deviation conditions above rule out exits from it. Thus it is the unique binary fixed point. □

5.4 Closed-loop interpretation

The fixed point is best read forward. If $z = 1$, then s_A is interpreted as a high signal, but this does not rescue the ambitious IC:

$$\lambda_A = 1 \Rightarrow \Delta_H(1, q) < 0 \Rightarrow z = 1 \text{ is not stable.}$$

If $z = 0$, then D1 maps the off-path deviation to low type:

$$\lambda_A = 0 \Rightarrow \Delta_H(0, q) < 0 \Rightarrow z = 0 \text{ is stable.}$$

High-type ambiguity therefore collapses itself: high-type withdrawal is both a current best response and the belief support for it. This is the belief-loop fixed point.

5.5 Scope of fixed point

This section is a pure-strategy fixed-point claim for the binary signal class. Interior mixed fixed points and lower-interval noise-induced mixing are intentionally out of scope here. Mixed and continuous-type refinements are treated in the polarization extension in Section 7.

6 Dynamic Forward Invariance

Section 5 establishes that in the binary class, the one-period object selects pooling for $q > q^*$. This section adds the missing microfounded dynamic state and asks whether endogenous accumulation can reverse that outcome.

6.1 Trust-capital law of motion

Let (C_t, E_t) be a two-dimensional endogenous state observed before j_t is realized.

- C_t is semantic trust capital created by repeated pooling-safe behavior.
- E_t is engagement reinforcement (habituation) accumulated from sustained non-ambitious actions.

The state evolution is

$$\begin{aligned}
C_{t+1} &= (1 - \rho_C)C_t + \eta_C \mathbf{1}\{s_t = s_P\} - \psi_C \mathbf{1}\{s_t = s_A\} - \xi_t^C, \\
E_{t+1} &= (1 - \rho_E)E_t + \eta_E \mathbf{1}\{s_t = s_P\} + \eta_A \mathbf{1}\{m_t = 1\} - \xi_t^E,
\end{aligned}$$

where $m_t \in \{0, 1\}$ is a cheap trust-restoring message available to both types and ξ_t^C, ξ_t^E are bounded asynchronous shocks. This keeps the mechanism endogenous while preserving one-period exposure at the signaling margin.

Define a temporary semantic penalty term

$$\Sigma_t = \omega_C C_t + \omega_E E_t \geq 0.$$

Then, in low-scrutiny state L, the low-state advantage of sending ambitious is

$$\Delta_\theta^{\text{dyn}}(q_t, \lambda_t) = B_\theta(\lambda_t) - \beta q_t \mu_H \Phi_\theta - \Sigma_t,$$

where the current posterior λ_t appears in B_θ as in Section 5. Sending s_P avoids the future repricing term and does not create today's semantic penalty.

6.2 Unpriced trust-restoring messages

Cheap anthropomorphic messages are unpriced: the sender can choose $m_t = 1$ with no direct signal-dependent cost and no type restriction. Because both types can choose m_t identically, the induced change in (C_t, E_t) is type-neutral. Therefore these messages are shadowable.

To see dynamic single crossing, let receiver response a induce continuation return $R(a)$ from choosing s_A. The high- and low-type off-path gains are

$$G_H(a;t) = R(a) - \bar{k}_H - \beta q_t \mu_H \Phi_H - \Sigma_t, \qquad G_L(a;t) = R(a) - \bar{k}_L - \beta q_t \mu_H \Phi_L - \Sigma_t.$$

The term Σ_t cancels in the comparison, so adding or relaxing trust-capital does not alter the type comparison. If $q_t > q_{\text{D1}}$, then $G_H(a;t) < G_L(a;t)$ for every receiver response, so s_A is still D1-attributed to the low type.

6.3 Dynamic Spence ratio and wedges

For any endogenous state (C_t, E_t) define the state-adjusted top-down ratio

$$\widetilde{q}_\theta^{\text{IC}}(t) = \frac{B_\theta(\lambda) + \Sigma_t}{\beta \mu_H \Phi_\theta}.$$

Endogenous trust-capital terms appear symmetrically in numerators and therefore do not change the fundamental type-ordering created by (Φ_H, Φ_L) and static primitives. They shift the threshold pair, but not the sign of the top-down reversal.

A true restoration requires an externally priced type-specific wedge, for example a direct penalty or premium that depends on sender type/claim context and is not a pure continuation by-product of C_t and E_t.

6.4 Forward invariance

Let

$$\mathcal{Q} := \{q \in [0,1] : q \geq q^*\}, \qquad q^* = \max\{q_{\text{IC}}, q_{\text{D1}}\}.$$

Assume that under pooling on s_P, the process satisfies:

1. $\Sigma_{t+1} - \Sigma_t \geq \underline{d} - \xi_t$ for some $\underline{d} > 0$ and bounded $\xi_t \in [0, \bar{\xi}]$;
2. $\Sigma_0 \geq 0$ and $\bar{\xi} < \underline{d}$;
3. $\beta q \mu_H \Phi_H - B_H(0) - \Sigma_0 > 0$ for all $q \in \mathcal{Q}$.

Theorem 3 (Pooling interval forward invariance)**.** *Under these assumptions and strict D1 preferences at $q \in \mathcal{Q}$, if period t is in the pooling assessment of Section 5, then period $t+1$ is also in that assessment with probability one. Equivalently, $\mathcal{Q}$ is almost surely forward-invariant.*

Proof. In the pooling assessment, $\lambda_t = 0$ for any off-path ambitious deviation. The high-type deviation margin is

$$\Delta_H^{\text{dyn}}(q_t, 0) = B_H(0) - \beta q_t \mu_H \Phi_H - \Sigma_t.$$

By assumption 3 and $\Sigma_t \geq \Sigma_0$, $\Delta_H^{\text{dyn}}(q_t, 0)$ is negative for all $q_t \in \mathcal{Q}$. Assumption 1 says endogenous drift in Σ_t dominates bounded asynchronous noise, so this negativity cannot be overturned by one-period shocks. Hence high type never has a profitable ambiguous-ambitious deviation from s_P once in the pooling band, and D1 therefore continues to map s_A to low attribution. The same argument applies period-by-period by induction. □

The conclusion is defensive. Dynamic trust-capital and habituation can be modeled explicitly and can affect local margins, but they do not create a new stopping phenomenon. They do not by themselves repair top-down collapse.

6.5 Population-density shape in the critical linearized system

For intuition (and to avoid stale wording from prior drafts), consider the linearized density

$$\mathcal{L}_\kappa \varphi = \lambda \varphi$$

for the trust-state law under pooling. The relevant object is its principal eigenpair (λ_1, φ_1); by definition $\varphi_1 \geq 0$ on the domain, not sign-changing. In the parameter regime of interest, the population profile is therefore single-peaked: it starts from the boundary with a left foot at $x = 0$, peaks in the safe region, and after the turning point decays into the dangerous region with exponential tail. A collapse-zone oscillation is not part of φ_1; that oscillatory sine pattern belongs to non-principal local approximations and is not a valid population density.

Airy appears only as a local normal form near the turning point, where the reduced equation takes the form

$$\phi''(x) - x\,\phi(x) = 0,$$

so $\phi(x) \propto \mathrm{Ai}(x)$ for matching. Elsewhere, φ_1 governs the full profile.

At $\lambda = 0$, the critical transition, imposing the Dirichlet condition at the left boundary yields the principal-root condition: the boundary is pinned at the first zero of the matching profile. This is not fine tuning; it is the algebraic consequence of the critical principal-eigenvalue condition together with the boundary constraint. The earlier "coincidence" wording should therefore be replaced by this inference.

7 Observable Implications: Polarization of Ambitious Signals

The binary model delivers a sharp equilibrium statement: once the transition probability into high scrutiny exceeds the critical threshold, the high type exits the ambitious signal and pooling on the safe signal is selected under D1-consistent beliefs. This section records the observable implication in a richer type space. Near the threshold, the quality of the ambitious signal need not decline smoothly. The more distinctive prediction is polarization: ordinary winners withdraw, while low-calibration agents and extreme high types may remain.

7.1 Continuous types

Let the type space be an interval

$$\Theta = [\underline{\theta}, \overline{\theta}].$$

The sender chooses between s_A and s_P. The current low-scrutiny net separation rent from s_A is $B(\theta)$. The true-type-indexed adverse-repricing exposure is $\Phi(\theta)$, derived from primitive self-accuracy levels

$$p(\theta) = \Pr(m = 1 \mid \theta), \qquad p'(\theta) > 0,$$

through

$$\Phi(\theta) = \alpha D_{\mathrm{KL}}\left(\mathrm{Bern}(p(\theta)) \,\|\, \mathrm{Bern}(\bar{p})\right).$$

The low-state payoff gain from the ambitious signal is

$$D(\theta, q) = B(\theta) - \beta q \mu_H \Phi(\theta).$$

Thus type θ chooses s_A exactly when

$$q < q^*(\theta), \qquad q^*(\theta) = \frac{B(\theta)}{\beta \mu_H \Phi(\theta)}.$$

The binary theorem corresponds to the case in which the top type's threshold is crossed first. In the continuous model, the entire shape of $q^*(\theta)$ determines which types remain on the ambitious signal.

7.2 Hollowing out

A simple mean-reversion story would predict that, as scrutiny risk rises, the ambitious signal gradually becomes lower quality. That is not the generic implication of top-down pooling. The model removes types according to the exposure-to-rent ratio

$$\frac{\Phi(\theta)}{B(\theta)},$$

not according to type itself.

Suppose $q^*(\theta)$ is nonmonotone: it is low on an intermediate-high region of the type space but rises again in the extreme upper tail. Economically, these are environments in which ordinary winners carry substantial reputation-at-risk but do not have enough surplus to dominate future adverse repricing, while extreme types have sufficiently large current surplus or sufficiently robust self-models to remain exposed.

For a given q, the set of types choosing the ambitious signal is

$$A(q) = \{\theta \in \Theta : q < q^*(\theta)\}.$$

If $q^*(\theta)$ has an interior trough, then for transition probabilities in the corresponding range,

$$A(q) = [\underline{\theta}, \theta_1(q)] \cup [\theta_2(q), \overline{\theta}]$$

for some $\theta_1(q) < \theta_2(q)$. The ambitious signal is then generated by two separated groups: low or poorly calibrated types whose exposure-to-rent ratio is low because they have little to lose, and extreme high types whose rents remain large enough to bear exposure.

Proposition 1 (Polarized degradation)**.** *Suppose $q^*(\theta)$ is continuous and there exist $\theta_1 < \theta_2$ such that*

$$q^*(\theta) > q \quad \textit{for} \quad \theta \in [\underline{\theta}, \theta_1] \cup [\theta_2, \overline{\theta}]$$

and

$$q^*(\theta) < q \quad \textit{for} \quad \theta \in (\theta_1, \theta_2).$$

Then the type distribution conditional on the ambitious signal has support on two separated intervals. In particular, the ambitious signal is not a monotone high-type signal in this region.

Proof. The sender chooses s_A if and only if $q < q^*(\theta)$. The stated inequalities imply that this condition holds exactly on $[\underline{\theta}, \theta_1] \cup [\theta_2, \overline{\theta}]$ and fails on (θ_1, θ_2). Hence the support of θ conditional on observing s_A is the union of two separated intervals. This support is not an upper set in Θ. Therefore s_A is not a monotone high-type signal. $\square$

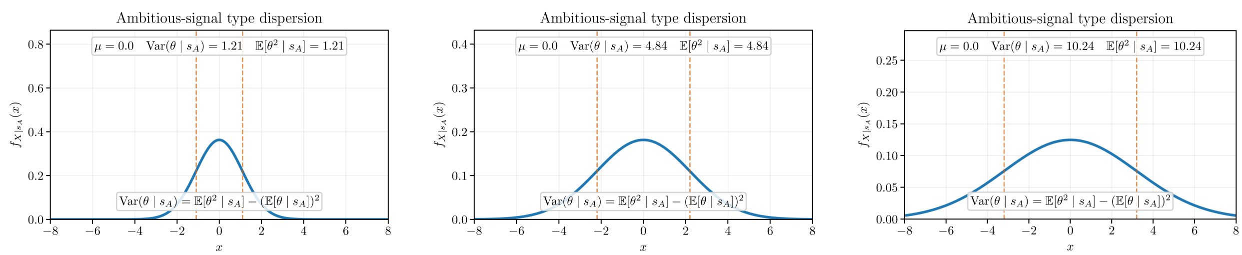
The observable shadow is a hollowing-out of credible ambition. The conditional mean of type among ambitious senders may fall,

$$\mathbb{E}[\theta \mid s_A] \downarrow,$$

but the more distinctive prediction is a rise in dispersion:

$$\mathrm{Var}(\theta \mid s_A) \uparrow$$

relative to the pre-threshold separating region. The surviving ambitious signal can be generated by extreme competence or by poor anticipatory discipline. This is why mean reversion is too weak a description. The ambitious signal becomes polarized.



$\mathrm{Var}(\theta \mid s_A) = 1.21$ $\quad$ $\mathrm{Var}(\theta \mid s_A) = 4.84$ $\quad$ $\mathrm{Var}(\theta \mid s_A) = 10.24$

Figure 1: Model simulations: increasing conditional dispersion for the ambitious signal.

7.3 Relation to the D1 flip

The polarization result is the observable analogue of the D1 flip in Section 4. In the binary model, once $q > q_{\mathrm{D1}}$, the off-path ambitious signal is attributed to the low type because the high type has the smaller profitable-deviation response set. In the continuous model, the same force makes the ambitious signal nonmonotone. It no longer means simply "high type." It can mean either extreme surplus or low anticipatory discipline.

The epistemic exposure term remains pinned to true type throughout this argument. The receiver's interpretation affects the reputational return to s_A, while $\Phi(\theta)$ describes the true-type cost of carrying exposed self-presentation into a possible adverse repricing state. This separation prevents the off-path belief system from redefining the underlying exposure technology.

7.4 Empirical language

The model predicts a specific kind of degradation. It is not merely that ambitious signals become noisier. It is that the middle of the credible ambitious class exits first. Agents with too little reputation-at-risk may continue because adverse repricing cannot take much from them. Agents with overwhelming surplus may continue because the ambitious signal remains worthwhile. Agents in between withdraw because they have enough to lose and not enough surplus to dominate the future exposure cost.

In the standard signaling model, the high signal is protected because bad types cannot climb up. In this model, the high signal is hollowed out because good types climb down. The remaining signal can look extreme: true outliers and badly calibrated imitators share the same visible action. That polarization is the observable signature of top-down pooling.

The polarization result identifies the cross-sectional shadow of withdrawal. What remains is to connect that shadow to the dynamic incentive that makes withdrawal rational before high scrutiny is realized. Section 6 supplies this step by showing that the same high-type IC threshold can be interpreted through forward invariance. Section 5 provides the fixed-point closure in the binary core.

References